

Project 4A - Wokwi Simulated Hardware Test Bench

Purpose: Simulate a microcontroller ADC test workflow using Wokwi and Python analysis.

Data file used: wokwi_serial_log.csv

Engineering Summary

Total samples: 48

ADC range: 0 to 1023

Voltage range: 0.000 V to 5.000 V

Average voltage: 2.607 V

Status counts: {'NORMAL': 38, 'FAIL': 7, 'WARN': 3}

Final result: FAIL

Pass / Warn / Fail Logic

NORMAL: voltage < 3.0 V

WARN: 3.0 V <= voltage < 4.0 V

FAIL: voltage >= 4.0 V

Conclusion

The simulated ADC workflow completed with final result: FAIL.

This proves the data path from analog-style input to ADC conversion, classification, CSV logging, plotting, and automated engineering reporting.

Voltage Plot

